

Santanu Mondal

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Summary

M.Tech graduate in **Network & Information Security** from Pondicherry University (2026), with research experience in blockchain, cryptography, and IoT security, including a peer-reviewed publication and several manuscripts under review. Seeking a **PhD position or research role** to continue contributing to research.

Education

Pondicherry University <i>M.Tech in Network & Information Security</i> Thesis: ZKP and PUF-Authenticated Dual-Offline CBDC Protocol with AML/CFT Compliance for Resource-Constrained IoT Devices Advisor: Dr. T. Chithralekha Coursework: Principles of Modern Cryptography, Network Security, Blockchain Technology, Distributed Systems & Security, Cyber Forensics, Machine Learning, Natural Language Processing, Information Security Management & Standards	Puducherry, India 2024–2026
South Asian University (A University Established by SAARC Nations) <i>M.Sc in Computer Science</i> Thesis: System for Managing Multiple Identities of IoT Devices using Blockchain [PDF] [Code] Advisor: Dr. Shachi Sharma Coursework: Cryptography & Network Security, Computer Networks, Advanced Data Structures & Algorithms, Data Mining	New Delhi, India 2021–2023
Asutosh College (University of Calcutta) <i>B.Sc (Hons.) in Computer Science</i> Project: Implemented k-means and KNN machine-learning algorithms from scratch in Python. [Code]	Kolkata, India 2018–2021

Publications

Published

1. S. Sharma, **S. Mondal**, S. Ishrat, "Managing Multiple Identities of IoT Devices Using Blockchain," *Soft Computing and Its Engineering Applications (icSoftComp 2023)*, CCIS 2031, Springer, 2024. DOI: 10.1007/978-3-031-53728-8_11

Preprints & Under Review

2. **S. Mondal** and T. Chithralekha, "Zero-Knowledge Proof (ZKP) Authentication for Offline CBDC Payment System Using IoT Devices," *Preprint*, arXiv:2603.03804. (Under review, journal)
3. **S. Mondal**, T. Chithralekha, "A Systematic Literature Survey of Cryptocurrencies," *Preprint*, ssrn.6290118. (Under review, journal)
4. **S. Mondal**, A. Paswan, "NLP-Based Browser Extension for Real-Time Malicious URL and Domain Detection," Presented at ICCET 2026. (Under review, journal)
5. **S. Mondal** and T. Chithralekha, "A Systematic Literature Review of Central Bank Digital Currencies (CBDCs): Socio-Economic Impact, Governance, and Global Adoption." (Under review)
6. **S. Mondal** and T. Chithralekha, "Technical Design Choices of Central Bank Digital Currencies (CBDCs): A Survey on Architecture, Ledger Technologies, and Consensus Mechanisms." (Under review)

Conference Presentations & In Preparation

7. **S. Mondal**, A. Paswan, "Revolutionizing Distance Learning: AI-Driven Personalization for the Next-Gen Education System," National Conference on Institutional Collaboration: A Pragmatic Approach for Transforming Distance Education, Pondicherry University & Loyola College Society, Chennai, April 2025.
8. **S. Mondal** and T. Chithralekha, "ZKP and PUF-Authenticated Dual-Offline CBDC Protocol with AML/CFT Compliance for Resource-Constrained IoT Devices." (Manuscript in preparation)

Projects

- **Multiple Identity Management of IoT Devices — Blockchain** [GitHub]
Designed a Hyperledger Fabric permissioned-blockchain system for managing multiple identities of IoT devices (M.Tech. thesis); implemented JavaScript chaincode (smart contracts) for device identity registration and asset transfer, built a client application on the Fabric Gateway SDK, simulated IoT device activity, and monitored the network with Hyperledger Explorer.
- **One-Human-One-Vote — ZK-Verified Anonymous Voting** [GitHub]
Built a privacy-preserving, decentralized voting dApp at the IIT Madras (CyStar) Winter School Hackathon; combined Solidity smart contracts with Merkle-tree eligibility commitments, zero-knowledge membership proofs, and nullifiers to guarantee one-human-one-vote anonymity without linking votes to voters.

- **Decentralized Voting DApp** [GitHub]
Developed an Ethereum voting smart contract (Solidity, Hardhat, Ethers.js) supporting multiple time-bound elections, one-vote-per-address enforcement, and real-time tallying; deployed and verified it on the Sepolia testnet via Alchemy RPC with event logging and admin access controls.
- **MalURLX — NLP-Based Malicious URL Detection** [GitHub]
Engineered an end-to-end ML system to detect malicious URLs from lexical features and character-level TF-IDF with a LightGBM classifier (~95–98% accuracy); served predictions through a FastAPI/Uvicorn REST API and shipped a companion Chrome extension for real-time browser protection.
- **Third Eye — Edge AI Assistive Navigation System** [GitHub]
Prototyped a fully on-device wearable navigation aid for visually impaired users on Arduino Nicla Vision at the ACM Winter School (IISc) Edge AI Hackathon; fused ToF distance sensing and IMU-based fall detection with an INT8-quantized vision model (10× smaller, 0.80 accuracy), streaming real-time alerts over UDP to a Flutter/Streamlit UI.
- **Alumni Management System — AWS Deployment** [GitHub]
Deployed a PHP/MySQL (LAMP) web application on an AWS EC2 Ubuntu instance; configured security groups and SSH access, and managed provisioning and database import via PuTTY, FileZilla, and phpMyAdmin.
- **Remote Desktop — Real-Time Screen Sharing & Control** [GitHub]
Built a Python remote-desktop tool that streams a host's screen and relays remote mouse/keyboard control; captured and JPEG-encoded frames with OpenCV, evolved the transport from TCP to TLS-secured and chunked-UDP streaming (~20 FPS), and replayed input events with pynput across multi-threaded clients.
- **Distributed Systems Lab** [GitHub]
Implemented a suite of distributed-computing programs, including Java RMI chat and client-server applications, DNS resolution, IP multicast, distributed image processing, and a distributed number generator, demonstrating remote invocation, group communication, and parallel-computing concepts.

Skills

- **Programming:** Python, C
- **Blockchain & Web3:** Ethereum, Smart Contracts, Hyperledger Fabric, Hardhat, Zero-Knowledge Proofs, Offline CBDC
- **Security & Compliance:** Cryptography, Security Auditing, Risk Management, ISO/IEC 27001, NIST, ISMS
- **Networking & Distributed Systems:** Socket Programming, TCP/UDP, Client–Server Architecture, Cisco Packet Tracer
- **AI & Data:** Machine Learning, NLP
- **Web, Cloud & Tools:** HTML, CSS, AWS EC2, GitHub, VS Code

Academic Activities & Conferences

1. **ACM Summer School 2022** IIT Delhi, Jul 2022
Organized by ACM India — Training on geometric modelling and computational methods. [Certificate]
2. **Paper Presentation (International Conference)** Dec 2023
5th International Conference on Soft Computing and its Engineering Applications. [Certificate]
3. **Paper Presentation (National Conference)** Apr 2025
Organized by Pondicherry University and Loyola College Society, Chennai. [Certificate]
4. **Winter School on Decentralized Trust & Blockchains 2025** IIT Madras, Dec 2025
Organized by CyStar — Focused on blockchain technologies, cryptography, distributed systems. [Certificate]
5. **ACM Winter School on Edge AI** IISc Bangalore, Jan 2026
Organized by ACM India — Intensive training on Edge AI, TinyML, Embedded AI, and IoT systems. [Certificate]
6. **Paper Presentation (International Conference)** Mar 2026
International Conference on Cognitive Informatics Engineering and Technology (ICCET 2026), Vidyaa Vikas College of Engineering and Technology, Tiruchengode, Tamil Nadu, India. [Certificate]